

YOUSEF AHMED

AI / ML Engineer

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate (GPA 3.62) with hands-on experience building and deploying ML/DL models across computer vision, classification, and data pipeline workflows. Published researcher who deployed real-time AI on edge hardware (NVIDIA Jetson Nano) achieving 96% accuracy at 20 FPS. Skilled in Python, PyTorch, TensorFlow, and Docker, with practical exposure to MLOps and cloud-adjacent tools.

EXPERIENCE

Machine Learning Engineer

Jun 2024 – Aug 2024

PioPetro • Ohio, United States (Remote)

- Completed a virtual training program on ML applications in the petroleum industry, covering Petroleum Exploration, Artificial Lift Technology, Well Intervention, and Well Stimulation.

AI/ML Intern

Jul 2023 – Sep 2023

Information Technology Institute (ITI) • Al Minufiyah, Egypt

- Developed and benchmarked supervised and unsupervised ML models (classification, regression, clustering) using Scikit-learn; applied cross-validation and performance metrics (Accuracy, F1, RMSE) across structured datasets.

AI/ML Intern

Aug 2023

Digital HUB (D-HUB) • Cairo, Egypt

- Built a breast cancer detection classifier achieving 95% accuracy; awarded 3rd place in a competitive AI challenge co-hosted with the Electronics Research Institute.

PROJECTS

Agri-Bot — AI-Powered Crop Disease Detection Robot (Graduation Project)

Nov 2023 – Jun 2024

- Designed a real-time computer vision pipeline using YOLOv8 and MobileNetV2 deployed on NVIDIA Jetson Nano, achieving 96% accuracy at 20 FPS with fuzzy PID motor control and a web interface

Personal RAG Chatbot

May 2025

- AI chatbot using LangChain, ChromaDB, and Groq (Llama 4 Scout), with a FastAPI backend and a custom HTML chat UI, enabling natural language Q&A over a structured personal knowledge base.

FMCG Data Engineering Pipeline

April 2025

- Built an end-to-end Lakehouse pipeline on Databricks using PySpark, Delta Lake, and Unity Catalog with Medallion Architecture (Bronze → Silver → Gold), Star Schema modeling, incremental loading, and a live Genie AI dashboard.

Ball Tracking Mobile Robot

May 2023

- Built a sensor-driven robot with a real-time vision inference pipeline, achieving 95% object detection accuracy and sub-5cm tracking error using autonomous navigation and control algorithms.

Steganography Image Detection

Feb 2023

- Trained a CNN-based classifier to detect hidden data embedded in digital images, achieving 90% classification accuracy.

EDUCATION

B.Sc. Mechatronics Systems Engineering

2020 – 2024

MSA University (supervised by University of Greenwich) • Giza, Egypt • GPA: 3.62 / 4.0

SKILLS

Programming: Python | C++ | MATLAB | SQL

ML / DL: Supervised & Unsupervised Learning | CNNs | RNNs | Transfer Learning | Object Detection (YOLO) | Ilms | RAG

Data Engineering: ETL/ELT | Medallion Architecture | Delta Lake | PySpark | Databricks | Star Schema

Frameworks: TensorFlow | PyTorch | Scikit-learn | Pandas | NumPy | FastAPI | OpenCV | LangChain | LangSmith | LangGraph

MLOps: Git | GitHub | Docker | ONNX Runtime | AWS (fundamentals)

Soft Skills: Problem-solving | Time management | Adaptability | Research

PUBLICATIONS

Agri-Bot: AI-Powered Crop Disease Detection and Medication Recommendation Robot

Published at the 8th International Undergraduate Research Conference (IUGRC 8) • 2024

CERTIFICATIONS

- Machine Learning Specialization — Coursera (2023)
- Supervised Machine Learning: Regression and Classification — Coursera (2023)
- Python 101 for Data Science — IBM Cognitive Class (2022)